

Observation Recency in Bernoulli Fusion for Intermittent Multirobot Tracking

Abstract—Intermittent contact between mobile robots brings together target beliefs with different observation histories. Weighting an entire Bernoulli density by direct-observation age can improve existence decisions while degrading spatial pooling. We propose existence recency (ER), a fusion rule for labeled multi-Bernoulli tracking that screens untouched priors, retains ordinary spatial weights, and applies observation recency to existence terms. Both detections and missed detections update the local observation age, which remains unchanged by relaying. A two-block variational formulation establishes that this weighting preserves the fused spatial density for fixed inputs. On nine public V2V4Real sequences with real per-vehicle detections, ER has lower mean OSPA than the evaluated MIL and track-consensus adaptations, but its age factor does not consistently improve its no-age ablation. Two eight-robot mechanism case studies, each with twenty Monte Carlo trials, show OSPA reductions of 3.1% and 24.7% relative to its no-recency ablation in split-rejoin and churn-departure scenarios. Common-target localization RMSE remains within 0.02% of that ablation, whereas applying recency to both blocks increases it by 16.8% and 3.1%. ER yields higher post-departure false-target cost; adding two-hit confirmation reduces false hypotheses at the cost of slower discovery. These results identify a tradeoff between timely existence decisions and spatial information retention in multirobot fusion.

I. INTRODUCTION

Mobile robots often acquire complementary views while moving between connected and disconnected regions. When two subteams meet again, a newly received track can be supported by recent local measurements, inherited from an earlier encounter, or still close to an unobserved birth prior. Conversely, a low existence probability can reflect a useful recent missed detection or merely a lack of sensing opportunity. These distinctions matter when the communication opportunity permits only a small number of fusion rounds. The information exchanged over a newly available link can therefore differ substantially in recency. Figure 1 summarizes the proposed fusion mechanism.

Random finite set filters provide a natural representation of uncertain target number and state. The labeled multi-Bernoulli (LMB) filter represents each candidate track by an existence probability and a conditional spatial density [1]. Geometric pooling, also called generalized covariance intersection or Kullback–Leibler averaging (KLA), is widely used when cross-correlations are unavailable [2]. Unequal fields of view (FoVs) create differences in label support and observation history. We ask: *where should direct-observation recency enter a Bernoulli fusion rule when robots reunite?*

Multiplying a source’s entire fusion weight by an age factor seems natural, but changes both existence and position pooling. A source that has not observed a track recently

can still carry a useful spatial estimate received from another robot. Reducing its spatial influence solely because its own sensor has been uninformative can therefore sacrifice localization while improving target-number estimates. We study an existence-only alternative: retain ordinary spatial weights on eligible inputs, and use direct-opportunity age to weight the Bernoulli existence terms while retaining the spatial overlap normalizer.

We make three contributions. First, we introduce existence recency (ER): an observation-history rule qualifies fusion inputs, and the time since a robot’s last direct sensing opportunity determines its existence weight. Second, we derive the resulting Bernoulli pool from a two-block variational formulation and establish its fixed-input spatial invariance. Third, we evaluate transfer to public two-vehicle real detections and use two eight-robot event families as mechanism case studies, with external fusion comparisons and ablations of qualification, recency placement, and confirmation.

II. RELATED WORK

A. Robot sensor fusion

Robot sensor fusion uses representations suited to the estimated state. The ROS robot_localization package provides generalized extended Kalman filter (EKF) fusion [3], while LIO-SAM organizes lidar and inertial constraints in a factor graph [4]. For dynamic-object tracking, Autoware combines data association with EKF state updates [5]. LMB representations also support robotic mapping: Gao *et al.* develop distributed multi-robot SLAM with LMB landmark maps and map fusion, evaluated on simulated and real data [6]. In our target-tracking setting, labeled Bernoullis represent object identity, existence, and state, while ER determines how robots combine their posteriors.

B. Cooperative robotic search and tracking

Dames studies distributed search and tracking with PHD filtering and robot motion [7]. Ramachandran *et al.* reconfigure a multirobot communication network after sensing-resource degradation, using a central optimization system [8]. DecSTER combines particle PHD filtering and Thompson-sampling action selection for decentralized search and tracking with unreliable, asynchronous inter-agent communication [9]. These works address decisions about where robots should move or how resources should be connected. We hold motion and attempted links fixed to isolate the effect of posterior fusion during intermittent contact.

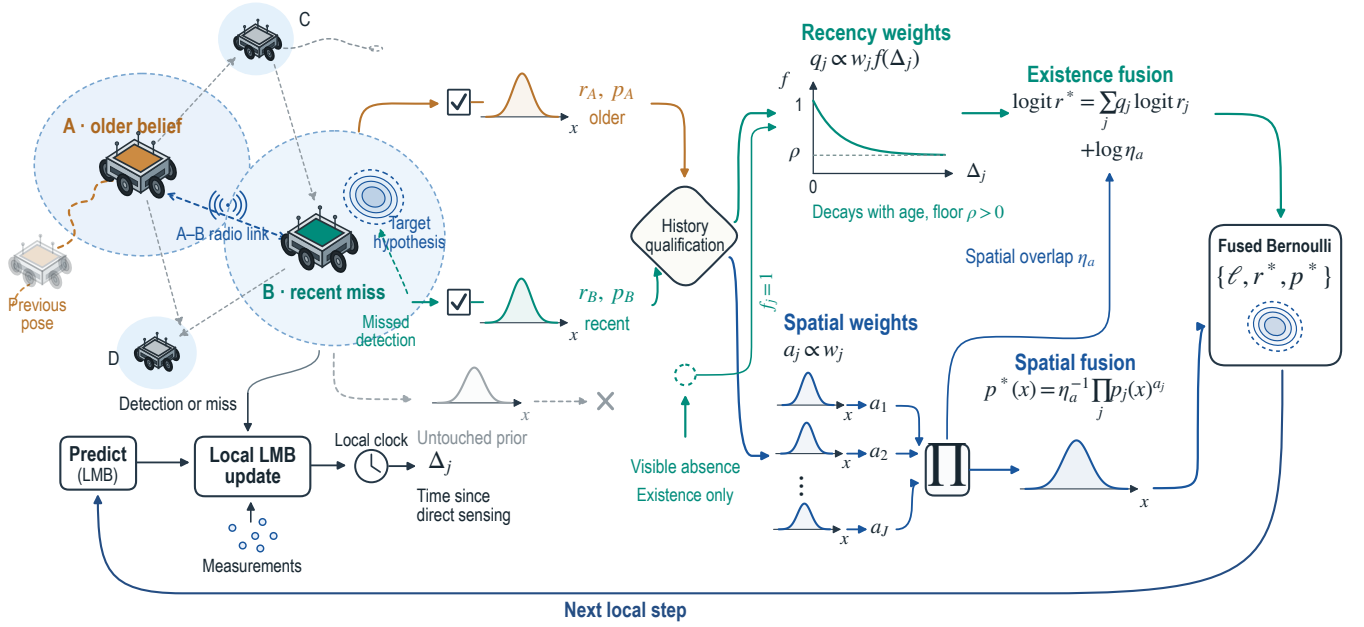


Fig. 1. Existence recency in multirobot tracking. Each robot predicts and updates an LMB, then exchanges posteriors at contact. Observation history qualifies fusion inputs; direct sensing age modulates existence weights q , while spatial pooling retains ordinary weights a . Visible absence contributes only existence evidence. The spatial-overlap normalizer η_a connects both branches, and the fused Bernoulli feeds the next local step.

C. Unequal views and information weighting

Wang *et al.* use information divergence to adapt LMB fusion weights across multiple views [10]. FoV compensation and multi-view geometric fusion are also established approaches [11], [12]. Gao *et al.* derive LMB-constrained minimum-information-loss (MIL) pooling and decompose label spaces for unequal views [13], [14]. Li *et al.* explicitly recognize that retained or relayed information can remain useful outside the current nominal FoV, and develop a density-decomposition CPHD approach [15]. Jin *et al.* examine shared-prior label consistency and introduce a consensus information selector [16]. Communication-aware adaptive KLA/LMB weights account for network constraints [17], while label assignment methods address inconsistent identities across nodes [18]. Nguyen *et al.* use finite-history trajectory consensus under limited FoV, maintaining local filtering independently of fused tracks [19]. We include this distinct fusion architecture explicitly. Building on these approaches, we examine where a direct-observation age factor should enter the Bernoulli fusion recursion.

D. Existence and spatial fusion

Üney *et al.* show that pointwise consistency of finite-set geometric mixtures does not ensure cardinality consistency, and advocate separate cardinality and localization fusion [20]. Our rule retains the spatial overlap penalty while assigning distinct weights to spatial densities and Bernoulli existence terms. This formulation lets us isolate the effect of observation recency on existence decisions and examine the resulting discovery–localization tradeoff at robot encounters.

III. PROBLEM AND FUSION RULE

A. Robot, sensing, and communication model

There are N robots with known poses $s_i(t)$ and a common clock. In the case studies, a physical edge exists within radio range and a geometry coordinator selects a bidirectional minimum spanning tree in each connected component. At each time, a robot predicts its LMB, updates with its own measurement set, and exchanges the resulting posterior for one synchronous fusion round. A dropped neighbor's Metropolis weight returns to the receiver's self weight. Fusion output is retained as the prior for the next local step. Communication cost includes coordinator traffic and lossy posterior messages.

A label ℓ identifies a Bernoulli after cross-source label alignment. For represented label ℓ , source j supplies $\pi_j^\ell = (r_j^\ell, p_j^\ell)$, where $r_j^\ell \in [0, 1]$ and $p_j^\ell(x)$ is a Gaussian-mixture density over position and velocity. The case studies use shared birth keys; the real-detection replay uses independent measurement-driven births and pairwise label assignment.

B. Observation history and participation

Each retained Bernoulli carries two metadata values. The history flag h_j^ℓ records whether the density descends from any scheduled local update with nonzero detection opportunity. A direct opportunity is evaluated on the predicted Gaussian means before the measurement update:

$$o_j^\ell(t) = 1 \left[\sum_g w_{jg}^\ell p_{D,j}(\mu_{jg}^\ell, t) > 0 \right]. \quad (1)$$

The local timestamp τ_j^ℓ is the last time this condition held at robot j . Both detections and missed detections count.

A received posterior propagates the history flag, while the receiver retains its own local timestamp after fusion. The resulting age measures time since the last *direct sensor opportunity*.

For each label, if at least one input carries observation history, represented inputs with no such history are excluded as untouched priors. Otherwise they retain ordinary participation. An absent label contributes a negative existence censor only when the represented spatial density places at least half its Gaussian-mean mass inside that source's current FoV, with expected detection at least 0.1. Its existence upper bound is the transmission threshold (0.01 in the case studies, 0.001 in the replay). Other absent sources abstain. Such censors are qualified negative evidence and are never assigned a spatial density. The flags propagate only through actual participating inputs. FoV occupancy is approximated at the Gaussian-component means.

Let $\mathcal{P}$ denote eligible represented inputs and $\mathcal{E}$ eligible existence inputs, including qualified absence censors. For the receiver's delivered Metropolis weights w_j , define

$$a_j = \frac{w_j \mathbf{1}[j \in \mathcal{P}]}{\sum_{k \in \mathcal{P}} w_k}, \quad q_j = \frac{w_j f_j \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} w_k f_k}. \quad (2)$$

For a represented label with a positive timestamp set

$$f_j = \rho + (1 - \rho) \exp[-(t - \tau_j^\ell) \Delta t / \lambda]. \quad (3)$$

Use $f_j = \rho$ for a represented but never-directly-observed label, and $f_j = 1$ for a qualified absence censor. We set $\rho = 0.25$ and $\lambda = 5$ s. The age factor is independent of the sign of the existence claim: a recent low r_j can outweigh an older high r_j . Positive self weight and the existing absence logic supply the ordinary single-input fallback; a label without eligible spatial support is omitted.

C. Existence recency with fixed spatial weights

For clarity suppress the label and receiver indices. Denote Bernoulli relative entropy by D_B and ordinary density relative entropy by D . For fixed eligibility and normalized a, q , consider

$$\min_{r \in (0,1), p} J(r, p) = \sum_{j \in \mathcal{E}} q_j D_B(r \| r_j) + r \sum_{j \in \mathcal{P}} a_j D(p \| p_j). \quad (4)$$

When the two supports and weight vectors agree, this reduces to the Bernoulli KLA objective.

Proposition 1 (fixed-input form). Assume $0 < r_j < 1$, $\eta_a = \int \prod_{j \in \mathcal{P}} p_j(x)^{a_j} dx > 0$, and finite relevant divergences. A minimizer of (4) is

$$p^*(x) = \eta_a^{-1} \prod_{j \in \mathcal{P}} p_j(x)^{a_j}, \quad (5)$$

$$\text{logit } r^* = \sum_{j \in \mathcal{E}} q_j \text{logit } r_j + \log \eta_a. \quad (6)$$

For fixed a , eligibility, and input densities, changing q changes existence without changing p^* .

Proof. Writing $p_a = \eta_a^{-1} \prod_{j \in \mathcal{P}} p_j^{a_j}$ gives $\sum_{j \in \mathcal{P}} a_j D(p \| p_j) = D(p \| p_a) - \log \eta_a$. For every $r > 0$ this is minimized at $p = p_a$. Differentiating the remaining scalar objective yields

$\text{logit } r - \sum_{j \in \mathcal{E}} q_j \text{logit } r_j - \log \eta_a = 0$. Its second derivative is $1/[r(1-r)] > 0$, establishing the stated solution. $\square$

For normalized densities, $0 < \eta_a \leq 1$, so the spatial overlap term remains a penalty in the existence update. Across recursive filtering steps, existence decisions can affect associations, pruning, and state extraction. We therefore evaluate localization on common assignment support alongside global set metrics.

The implementation calls the existing mixture-aware Gaussian-power fusion routine with a for spatial pooling and q for existence. It retains at most three components per input for the powered-mixture calculation, at most 256 component tuples, and at most eight output components. A non-admissible approximate overlap normalizer triggers a moment-matched fallback. This provides a finite-mixture approximation of the weighted density product. The case studies use this mixture cap; the real replay uses one moment-matched Gaussian per Bernoulli. Density methods share prediction, measurement update, and MAP-cardinality extraction within each evaluation. We call the rule *existence recency* (ER); using q in both blocks is the distinct *ER (both)* ablation.

D. Confirmation ablation and communication cost

A separate ablation delays history-based input restriction until some input carries a positive confirmation. A robot creates this bit after two consecutive local updates with opportunity and detection-association mass at least 0.5. Empty measurements reset the local streak. The confirmation bit propagates through fusion, while the hit counter and observation timestamp remain local. Ordinary FoV-aware pooling applies before confirmation. This two-hit rule tests the tradeoff between false-track rejection and discovery during short sensing contacts.

Density methods transmit the retained Gaussian parameters, labels, and metadata; trajectory methods transmit local track histories. Each directed link permits one packet per scan, divided into 16,384-byte fragments as needed, with a modeled 128-byte-per-robot control charge. We report raw payloads and fragment costs separately. Shared geometry is an input whose transport cost is excluded. ER adds constant metadata per represented label and linear-time weight calculations.

IV. EXPERIMENTS

A. Public real-detection replay

We use all nine evaluation sequences released with DM-STrack [21] for V2V4Real [22]: 1,993 paired frames at 10 Hz from two vehicles. Inputs are each vehicle's released no-fusion neural detections, per-frame relative transforms, and car labels. The detections already use a confidence cutoff of 0.20; no additional score threshold is applied. Ground truth supplies evaluation targets only. Births and cross-source associations use detections and posterior densities.

We evaluate planar car centers within 40 m of either sensor and inside the annotated ego-frame rectangle $[-70.4, 70.4] \times [-40, 40]$ m. A 3 m disk around each sensor excludes the

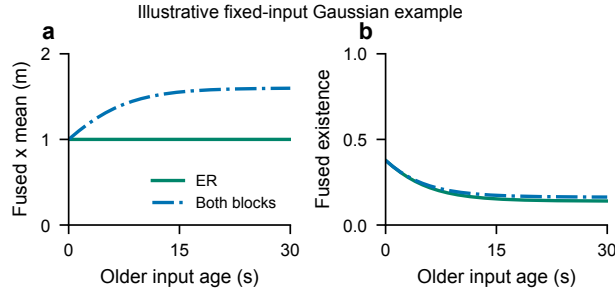


Fig. 2. Effect of observation age in a two-source Gaussian example. Two unit-covariance Gaussian densities have x means 0 and 2 m, with existence 0.9 at the older source and 0.1 at the current source. Ordinary spatial weights are equal. Increasing the older source’s direct age changes ER’s existence while leaving its spatial mean at 1 m. Applying age to both blocks also moves the spatial mean. Both rules retain their respective spatial overlap normalizer.

sensing platforms, whose bodies are masked by detector preprocessing but present in the labels. The same domain applies to detections, reported estimates, and truth, and to the likelihood and absence tests. The 40 m crop defines a conservative modeled FoV, not an occlusion map. Detections are already expressed in the current ego frame; the second vehicle’s transform locates its sensor. The relative-frame constant-velocity model lacks ego-motion compensation.

The shared LBP-LMB backend moment-projects each Bernoulli to one Gaussian, including association uncertainty. It uses $\Delta t = 0.1$ s, survival 0.99, white-acceleration intensity $25 \text{ m}^2/\text{s}^3$, observation covariance $I_2 \text{ m}^2$, nominal in-domain $p_D = 0.9$, Poisson clutter mean three per scan, and pruning at $r \leq 0.001$. Each source independently creates a birth at every previous-scan detection with $r_B = 0.01$, zero velocity, and covariance $\text{diag}(16, 16, 225, 225)$. These fixed model assumptions are shared by all methods. ER retains $\rho = 0.25$, $\lambda = 5$ s.

Each sequence is replayed with reliable links and with emulated intermittent links: independent directed loss of 0.1 plus a bidirectional outage over the middle 20% of scans. All methods share the delivered graph and one packet opportunity per direction per scan. Received density pairs use equal weights. This is an exploratory two-dimensional fusion evaluation: preflight domain and resource corrections preceded the complete cohort. It is not the official three-dimensional benchmark or a reproduction of DMSTrack’s learned tracker.

B. Mechanism case studies

Eight simulated robots follow two smooth patrol families for 120 frames at $\Delta t = 0.5$ s. Their maximum speed is below 3 m/s and acceleration below 2.1 m/s^2 . Radio range is 24 m and circular sensing range is 14 m. The fixed paths induce connected and two-component periods (Fig. 3). All targets use a constant-velocity state with independent per-step acceleration standard deviation 0.016 m/s^2 . The filter uses a continuous white-acceleration covariance with intensity 0.0128 and survival probability 0.995. Within FoV,

detection probability is 0.9 and Cartesian observation noise has standard deviation 0.4 m. Poisson clutter has mean one per robot per frame and is uniform over a fixed 104×80 m rectangle. Local pruning discards existence at or below 0.01; belief propagation uses tolerance 10^{-6} and at most 1,000 iterations. All methods share these settings.

The *split-rejoin* scene has two initial targets and two targets born at frame 35 while the teams are separated. The *churn-departure* scene uses repeated contact changes and removes one late target after frame 90; the filters receive no departure notification. In the *no-new* control, only the two initial targets exist, although identical later birth priors are still offered. Six fixed candidate regions have prior existence 0.1, position standard deviation 3.2 m, and velocity standard deviation 0.2 m/s. Regions 1–2 are activated at frame 1 and regions 3–6 at frame 35. Regions 5–6 remain empty in every scene. The filters receive the candidate birth schedule and priors.

Each scenario uses twenty Monte Carlo trials. In each trial, every robot receives an independent constant path offset drawn uniformly from $[-0.6, 0.6]^2 \text{ m}$, and the communication components and trees are recomputed. The offsets vary the layout and contacts while preserving speed and acceleration. Target trajectories, detections, clutter, and directed link-loss draws are paired across methods. Directed packet loss is 0.1.

C. External methods, controls, and ablations

*TC-OSPA*² uses the unchanged author kinematic matching and multi-neighbor fusion functions [19], with windows of five and ten scans, matching order one and cutoff 12 m. Its input is the shared Local backend’s track estimates with independent node label namespaces. Following the author’s architecture, fused tracks do not feed back into local densities. Both kinematic fusion stages are local computations on communicated histories. We evaluate set estimates; network-wide reporting label reconciliation is excluded. This adapts the published strategy to our local backend rather than reproducing its original filtering experiment.

MIL-AM implements the public Gao author manuscript’s augmented label assignment and common/exclusive label-subspace LMB-MIL [23]. For Gaussian inputs, matching uses its symmetric KL cost, with unmatched cost 50 per side. Common labels use arithmetic existence pooling and existence-weighted spatial mixtures; exclusive labels retain their source density. The same label alignment is applied to ER and its no-age ablation. Spatial mixtures are moment-projected in the real replay. This implementation is tied to the public manuscript; equivalence to the complete TAES 2022 version [14] is not established.

Local runs without communication. *FoV* uses FoV-based absence sensors. *ER w/o age* adds observation-history qualification and omits the age factor. *Age-all* applies direct-age weights to both spatial and existence pooling without history-based qualification. *ER (both)* adds this qualification, separating placement of the age factor from input selection.

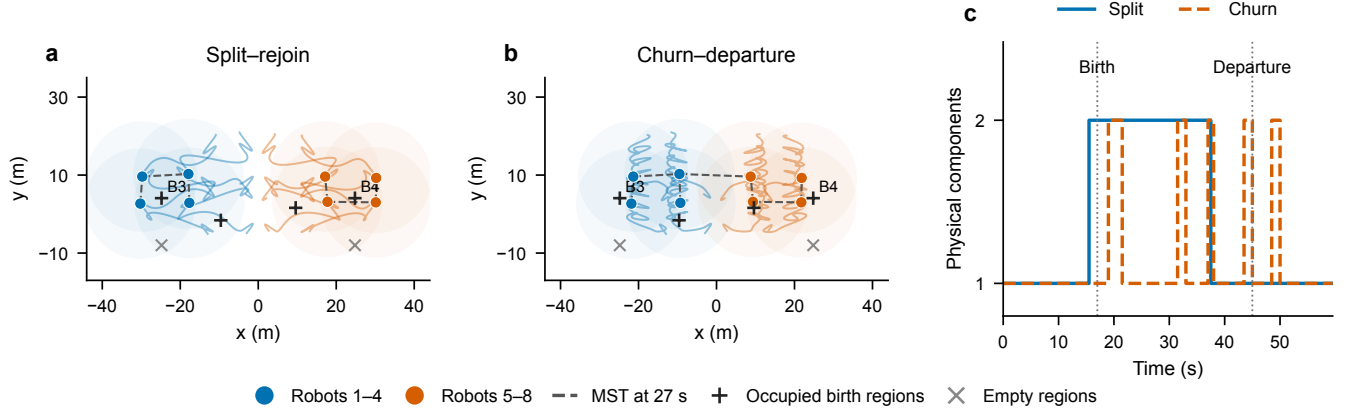


Fig. 3. Mechanism case studies. Thin curves are robot paths; circles and dashed tree edges show the state at 27 s. Light disks denote 14 m FoVs. Crosses mark candidate birth regions; the two lower regions remain empty. The split and churn scenarios create different contact histories, with target departure at 45 s in the churn scenario.

ER uses the same qualification with age in existence alone. *Confirmed ER* adds the two-hit gate.

The *MIL-Z* control uses constrained LMB MIL, $r = \sum_j w_j r_j$ and $p = \sum_j w_j r_j p_j / r$, with zero existence for absent labels [13]. *MIL-S* specializes label-subspace MIL [13], [14] to the shared-label setting: for each subspace defined by a source-membership pattern, weights are renormalized over represented sources before applying the same pooling formula. Both MIL controls use the shared birth keys for label alignment and retain untouched represented priors. Their Gaussian-mixture unions are reduced to the common component cap.

MIL-AM is evaluated on the real replay with independent births. The shared-label *MIL-Z/S* controls and the additional mechanism ablations are evaluated in the robot case studies.

D. Metrics and uncertainty

We report position OSPA with order 2 and cutoff 12 m [24], cardinality absolute error, and GOSPA with $p = 2, c = 12 \text{ m}, \alpha = 2$ [25]. Localization, missed-target, and false-target GOSPA components are reported as *squared costs* in m^2 . In the case studies, all extant targets are scored at every robot, including during disconnection. The scores therefore measure team-wide information coverage as well as local tracking.

For localization comparisons, assignments are computed separately for each method and matched errors are pooled only on robot-time-truth triples matched within the cutoff by both methods. We report the resulting RMSE and common-support size; unmatched targets contribute to OSPA/GOSPA. Departure false cost uses frames 91–120. Reunion OSPA uses the next ten frames, truncated at a further disconnection. For opposite-subteam robots, acquisition is defined geometrically as the first of three consecutive assignments within 2 m of a late target, censored at that target’s departure or the trial end.

We use 10,000 paired bootstrap resamples for 95% percentile intervals. The unit is a complete trial for the case

studies and a complete sequence for real data; sequence means receive equal weight. Frames and robots are not independent replicates. With only nine potentially related driving sequences, real-data intervals are descriptive.

V. RESULTS

A. Transfer to real detections

Table I reports all nine sequences. With reliable links, ER’s macro-average OSPA is 4.875 m, compared with 5.177 m for MIL-AM and 5.690/5.574 m for TC windows five/ten. With intermittent links, the corresponding values are 5.119, 5.309, and 5.792/5.765 m. ER beats MIL-AM in seven of nine reliable-link sequences and eight of nine intermittent-link sequences, and each TC window in eight of nine sequences under either condition. Sequence 0004 favors MIL-AM and TC over ER.

The age factor itself does not yield a consistent transfer gain. Relative to ER w/o age, paired OSPA differences are +0.162 m [-0.092, 0.517] with reliable links and +0.011 m [-0.077, 0.136] with intermittent links. Both intervals include zero. ER improves OSPA in five of nine sequences under each condition, but increases mean false-target cost by 6.73 and 4.46 m^2 , respectively. Thus the external-method comparison supports the full fusion configuration in this replay; it does not isolate a benefit of recency. Figure 4 retains every sequence comparison.

B. Case studies: existence recency and set estimation

Tables II and III separate external/control comparisons from mechanism ablations. Relative to ER without recency, ER reduces mean OSPA from 2.651 to 2.569 m in split-rejoin and from 1.123 to 0.846 m in churn-departure. The paired differences are -0.082 m [-0.110, -0.051] and -0.277 m [-0.399, -0.159], respectively. In the no-new control, the difference is -0.129 m [-0.167, -0.095]. Brackets denote the 95% bootstrap intervals defined above. Cardinality error decreases in all three scenarios. In the reunion windows,

TABLE I

V2V4REAL REPLAY OVER NINE SEQUENCES (1,993 FRAMES). OSPA: EQUAL-WEIGHT MEAN $\pm$ SEQUENCE SD (M); COUNT: CARDINALITY MAE; FALSE: GOSPA SQUARED FALSE-TARGET COST (M²).

Method	Reliable links			Intermittent links		
	OSPA	Count	False	OSPA	Count	False
Local	6.575 ± 1.928	2.227	73.86	6.575 ± 1.928	2.227	73.86
MIL-AM	5.177 ± 1.905	1.361	92.13	5.309 ± 1.949	1.457	94.35
TC-OSPA ² (5)	5.690 ± 2.185	1.489	115.12	5.792 ± 1.815	1.704	103.60
TC-OSPA ² (10)	5.574 ± 2.047	1.448	101.10	5.765 ± 1.760	1.700	93.09
ER	4.875 ± 1.868	1.291	83.97	5.119 ± 1.870	1.375	85.57
ER w/o age (ablation)	4.713 ± 1.755	1.316	77.24	5.109 ± 1.809	1.402	81.11

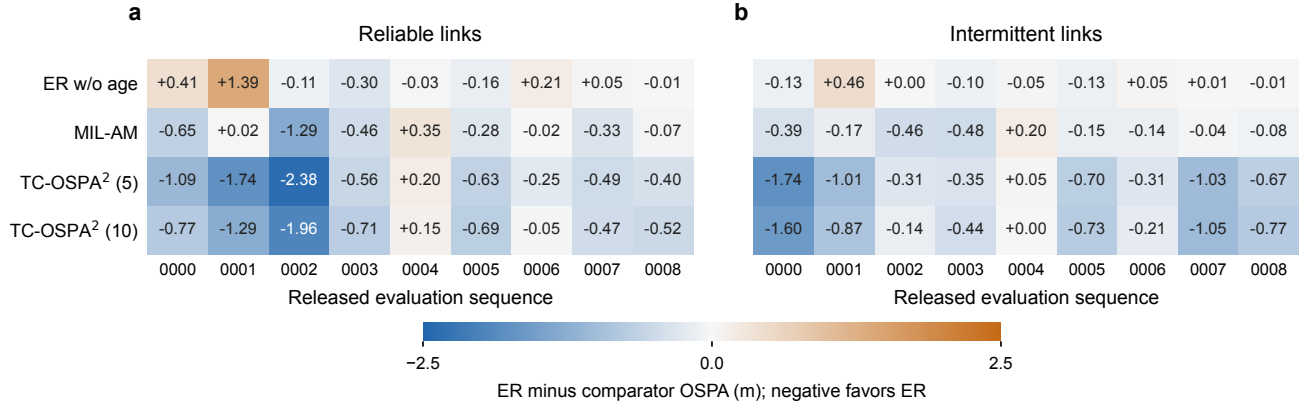


Fig. 4. Real-detection replay: ER minus comparator OSPA for each released evaluation sequence. Blue/negative values favor ER; orange/positive values favor the comparator. All nine sequences and both radio conditions are shown with one color scale. Cells are sequence summaries, not independent significance tests.

TABLE II

MECHANISM CASE STUDIES AND NO-NEW CONTROL: TWENTY PAIRED TRIALS EACH. OSPA IS MEAN $\pm$ TRIAL SD (M); COUNT IS CARDINALITY MAE. TC USES LOCAL TRACK HISTORIES WITHOUT DENSITY FEEDBACK; MIL-Z/S USE SHARED LABEL KEYS.

Method	Split-rejoin		Churn-departure		No-new control	
	OSPA	Count	OSPA	Count	OSPA	Count
Local	7.099 ± 0.454	1.368	7.280 ± 0.537	1.308	6.004 ± 0.839	0.645
FoV	3.713 ± 0.584	0.614	2.867 ± 0.551	0.442	0.804 ± 0.163	0.056
MIL-Z	4.812 ± 0.499	0.641	4.204 ± 0.574	0.452	2.765 ± 0.854	0.032
MIL-S	4.762 ± 0.506	0.624	4.112 ± 0.582	0.431	2.769 ± 0.854	0.033
TC-OSPA ² (5)	5.050 ± 0.491	0.861	5.538 ± 0.648	0.857	3.532 ± 0.998	0.305
TC-OSPA ² (10)	5.096 ± 0.496	0.867	5.598 ± 0.655	0.870	3.578 ± 1.000	0.308
ER	2.569 ± 0.110	0.398	0.846 ± 0.205	0.103	0.668 ± 0.219	0.040

TABLE III

INTERNAL MECHANISM ABLATIONS. OSPA (M) IS AVERAGED OVER TWENTY TRIALS. RMSE RATIOS USE COMMON ASSIGNMENT SUPPORT RELATIVE TO ER W/O AGE; FALSE COSTS ARE SQUARED GOSPA COMPONENTS (M²).

Method	Split	OSPA		Common-target RMSE ratio		False cost	
		Churn	No-new	Split	Churn	Departure	No-new
ER w/o age	2.651	1.123	0.797	1.0000	1.0000	10.785	0.679
Age-all	2.912	1.960	0.745	—	—	9.690	0.004
ER (both)	2.589	0.937	0.752	1.1685	1.0307	10.905	0.570
Confirmed ER	2.607	1.519	0.652	1.0018	1.0028	10.815	0.004
ER	2.569	0.846	0.668	0.9999	0.9998	12.255	0.570

OSPA is 2.533 versus 3.264 m in split-rejoin and 0.926 versus 1.248 m in churn-departure (ER versus ER w/o age).

FoV, MIL-Z/S, and both TC windows have higher OSPA than ER in these case studies. TC uses independent local histories and no density feedback, whereas recursive density fusion relays information over successive tree edges. Since every robot is scored against the global target set, this architectural difference contributes to the large TC gap; it cannot be attributed to the age factor. Support partitioning improves MIL in the two event cases. Local has the largest OSPA, consistent with its limited coverage.

C. Where the age factor belongs

Table III and Fig. 5(b) distinguish existence gains from localization. With the same observation-history qualification, ER (both) increases common-target RMSE relative to ER w/o age by 16.8% in split-rejoin and 3.1% in churn-departure. ER leaves this RMSE almost unchanged, with ratios of 0.9999 and 0.9998. This agrees with the motivation in Fig. 2: recent direct sensor opportunity is useful for existence weighting, while a posterior also retains spatial information from earlier observations and robot encounters.

D. False hypotheses and the confirmation tradeoff

Table III shows the false-hypothesis tradeoff. ER's mean post-departure false cost is 12.255 m², compared with 10.785 m² for ER w/o age. The paired difference is +1.470 m² [−0.705, 3.795]. Similarly, ER's no-new false cost is 0.570 m², versus 0.00375 m² for FoV. Lower OSPA therefore does not imply better false-track rejection.

Confirmed ER reduces no-new false cost to 0.00375 m², but its churn OSPA is 1.519 m, versus ER's 0.846 m. This is consistent with the cost of requiring consecutive local positive associations during short sensing opportunities. In churn, ER acquires 160/160 opposite-subteam robot-target queries, while Confirmed ER acquires 152/160. Among successful acquisitions, mean delays are 2.47 and 7.63 s, respectively. The remaining eight Confirmed ER queries are unacquired by their censoring time. Discovery and post-departure persistence therefore require separate metrics.

E. Communication cost

In the case studies, all communicating methods use the same attempted and delivered messages for each paired trial. Mean attempted wire cost is 24.98 MiB for split-rejoin and no-new, and 25.74 MiB for churn-departure. Across communicating methods and scenarios, mean unpadded payload totals range from 2.14 to 9.85 MiB; the padded message budget is identical. TC's raw payloads are 0.68–1.76 MiB. In the real replay, equal packet opportunities yield different fragment costs: mean attempted wire volume per sequence is 8.22/8.24 MiB for ER, 9.97/9.85 MiB for MIL-AM, and 6.97/6.97 MiB for either TC window (reliable/intermittent). ER's raw payloads are 4.35/4.48 MiB, compared with 1.35/2.64 MiB for TC windows five/ten. This is not an equal-byte comparison.

VI. DISCUSSION AND CONCLUSION

ER uses observation history to select participating inputs and direct sensing age to weight existence, while retaining ordinary spatial weights. In the robot case studies, this separation reduces set error relative to ER without recency. In split-rejoin and churn-departure, it also avoids the common-target localization degradation of applying age to both blocks. A robot's direct sensing age and the spatial information accumulated in its posterior describe different properties: earlier observations and relayed evidence can remain useful after direct contact ends.

False-track rejection introduces a further tradeoff. ER retains some false hypotheses, and the two-hit confirmation gate reduces no-new false cost at the expense of discovery during short sensing contacts. The departure and acquisition results show why set error, localization, false cost, and discovery delay should be evaluated together.

The real-detection replay broadens the evidence beyond these prescribed events and removes shared truth-informed birth schedules. It also exposes a transfer limit: the complete ER configuration improves on the evaluated external adaptations, but its age term adds no consistent OSPA gain over the no-age ablation and increases false cost. The case-study effect should therefore be interpreted as conditional on sensing and contact history.

The replay uses two vehicles, a cropped planar task, approximate Gaussian detector likelihoods, moving ego coordinates without motion compensation, and emulated radio loss. It establishes neither full 3-D benchmark ranking nor performance in a larger real robot network. Label matching and detector calibration also affect existence decisions. Improving that calibration and testing contact-dependent confirmation are concrete next steps before claiming a broadly transferable recency benefit.

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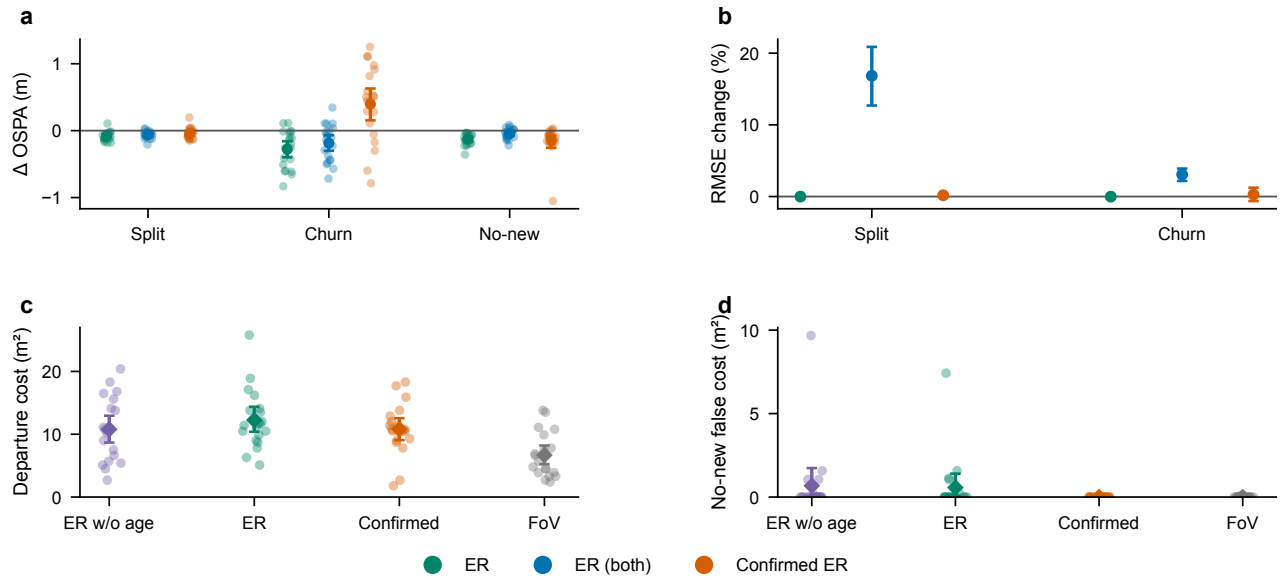


Fig. 5. Set error, localization, and false-target costs. (a) Paired OSPA differences relative to ER w/o age; small dots show individual trials, and large points with bars show means and 95% bootstrap intervals. (b) Percentage changes in pooled RMSE on common assignment support. (c,d) Departure and no-new false-target squared costs, with twenty trial points per method. ER reduces set error with little change in common-target localization; confirmation changes the discovery–false-cost tradeoff.

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